

4 (a)	charge / potential (difference) (<i>ratio must be clear</i>)	B1	[1]
(b) (i)	$V = Q / 4\pi\epsilon_0 r$	B1	[1]
(ii)	$C = Q / V = 4\pi\epsilon_0 r$ and <u>$4\pi\epsilon_0$ is constant</u> so $C \propto r$	M1 A0	[1]
(c) (i)	$r = C / 4\pi\epsilon_0$ $r = (6.8 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12})$ $= 6.1 \times 10^{-2} \text{ m}$	C1 C1 A1	[3]
(ii)	$Q = CV = 6.8 \times 10^{-12} \times 220$ $= 1.5 \times 10^{-9} \text{ C}$	A1	[1]
(d) (i)	$V = Q/C = (1.5 \times 10^{-9}) / (18 \times 10^{-12})$ $= 83 \text{ V}$	A1	[1]
(ii)	<i>either</i> energy = $\frac{1}{2}CV^2$ $\Delta E = \frac{1}{2} \times 6.8 \times 10^{-12} \times 220^2 - \frac{1}{2} \times 18 \times 10^{-12} \times 83^2$ $= 1.65 \times 10^{-7} - 6.2 \times 10^{-8}$ $= 1.03 \times 10^{-7} \text{ J}$ <i>or</i> energy = $\frac{1}{2}QV$ $\Delta E = \frac{1}{2} \times 1.5 \times 10^{-9} \times 220 - \frac{1}{2} \times 1.5 \times 10^{-9} \times 83$ $= 1.03 \times 10^{-7} \text{ J}$	C1 C1 A1 (C1) (C1) (A1)	[3]

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5 (a) field into (the plane of) the paper

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